Cédric Bonhomme

Information security, vulnerability management & open-source software

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Profile

Computer scientist working in information security within a Computer Emergency Response Team (CIRCL, Luxembourg). I design and maintain open-source security software, most notably Vulnerability-Lookup, and conduct applied research on vulnerability management: Coordinated Vulnerability Disclosure, severity classification with language models, and forecasting of vulnerability sightings. I enjoy sharing this work through papers, talks and trainings.

Experience

Luxembourg House of Cybersecurity

2017 – present

CIRCL — Computer Incident Response Center Luxembourg Luxembourg

Member of the CIRCL team (formerly part of SECURITYMADEIN.LU), involved in the activities of a national CERT: incident response, open-source tooling, research and training.

- **Vulnerability-Lookup** — main developer of an open-source platform that aggregates vulnerability data from many sources (CVE, GCVE, CSAF, security advisories, …), correlates it with sightings collected from the Fediverse and other feeds, and supports Coordinated Vulnerability Disclosure workflows.
- **GCVE** — contributions to the Global CVE allocation system and its tooling (github.com/gcve-eu).
- **Machine learning for vulnerabilities** — VLAI, a RoBERTa-based model for automated severity classification published on Hugging Face; models forecasting sparse and bursty vulnerability sightings; mapping CVEs to MITRE ATT&CK techniques.
- **Risk management** — development of MONARC (optimised risk assessment method), MOSP (collaborative JSON object platform) and Fit4Cybersecurity.
- **MISP** — contributions to the open-source threat-intelligence sharing platform.
- Numerous trainings and talks, on site and remote (FIRST, Hack.lu, CSAF Community Days, …).

Luxembourg Institute of Science and Technology

2009 – 2017

R&D engineer (formerly Public Research Centre Henri Tudor) Luxembourg

- **Android reverse engineering** — visualisation tool for malware identification, with model injection; the metamodel is built from a Protobuf schema using a PEG.js grammar.
- **Open Shelter Database** — platform collecting and sharing information on humanitarian shelters, developed for the Netherlands Red Cross (github.com/rodekruis/shelter-database).
- **Technology watch** — SaaS technology-watch solution built with Flask, MongoDB and Gevent.
- **BUGYO Beyond (EU project)** — analysis of the evolving security policies of critical infrastructures, scenarios for measuring security assurance levels, implementation with the JADE multi-agent framework.
- **WiSafeCar** — analysis and design of a solution based on multi-agent systems with SPADE (Smart Python Agent Development Environment).
- **MICIE (EU project)** — support tools for real-time risk prediction in interdependent critical infrastructures.

Selected open-source projects

Hosted on GitHub and SourceHut; a fuller list is available at cedricbonhomme.org/software.

- **Vulnerability-Lookup** — unified multi-source vulnerability data and CVD workflows.
- **Newspipe** — self-hosted web news aggregator.
- **Stegano** — pure-Python steganography module.
- **pyHIDS** — host-based intrusion detection system.
- **Iterated Function Systems** — exploration of IFS fractals in OCaml.
- **Freshermeat** — open-source software directory and release tracker.

Skills

Programming	Python, OCaml, C, Bash (advanced); JavaScript, C++, Go, Rust (good); Java, PHP, Perl, Haskell (notions).
Data & ML	Hugging Face Transformers, language-model fine-tuning, forecasting of count time series.
Security	vulnerability management and CVD, CVE/CSAF/CVSS ecosystems, threat intelligence (MISP), cryptography (RSA, ElGamal, PGP, PKI), network protocols and security (TCP/IP, DNS, BGP, OSPF).

Risk management	ISO/IEC 2700x, ISO/IEC 15408 (Common Criteria), MONARC.
Systems	GNU/Linux, OpenBSD; virtualisation with KVM, Xen and VMware ESXi.
Writing	LaTeX, HTML, Markdown; scientific papers and technical documentation.

Education

Master SSIC (information systems security), *Université Paul Verlaine, Metz* 2008 – 2009
Master 1 in Computer Science, *Université Paul Verlaine, Metz* 2007 – 2008
Bachelor's degree in Computer Science, *Université Paul Verlaine, Metz* 2006 – 2007
DEUG MIAS (mathematics, computer science, physics), *Université Paul Verlaine, Metz* 2002 – 2005
Baccalauréat S (mathematics, physics), *Sarreguemines* 2002

Languages & interests

Languages	French (native), English (fluent: read, written and spoken), German (intermediate understanding).
Interests	photography, piano, running, fractals and mathematics.

Publications

Also listed on cedricbonhomme.org/publications, arXiv and ORCID.

2026

- L. Bernardy, **C. Bonhomme**, A. Thirion, A. Dulaunoy, J. Samhi, T. F. Bissyandé, J. Klein, *SoK: Ten Years of Dark Jargon Detection and Interpretation from an NLP Perspective*. 42nd IEEE Annual Computer Security Applications Conference (ACSAC 2026), Los Angeles, CA, USA.
- **C. Bonhomme**, A. Dulaunoy, *Mapping CVEs to MITRE ATT&CK Techniques: A Curated Gold-Set Classifier and the Limits of LLM-Assisted Label Expansion*. Preprint, July 2026. arXiv:2607.25572.
- **C. Bonhomme**, *Building New IFS Attractors: a Working Vocabulary of Affine Maps*. July 2026. doi:10.5281/zenodo.21804778.
- **C. Bonhomme**, A. Dulaunoy, *Borrowing Strength Across Vulnerabilities: Hierarchical and Type-Aware Forecasting of Sparse Sightings*. June 2026. doi:10.5281/zenodo.22023077.
- **C. Bonhomme**, A. Dulaunoy, *Modeling Sparse and Bursty Vulnerability Sightings: Forecasting Under Data Constraints*. FIRST Cyber Threat Intelligence Conference 2026, Munich, Germany. arXiv:2604.16038.

2025

- **C. Bonhomme**, A. Dulaunoy, *VLAI: A RoBERTa-Based Model for Automated Vulnerability Severity Classification*. 25V4C-TC: 2025 Vulnerability Forecasting Technical Colloquium, Darwin College, Cambridge, UK, September 2025. arXiv:2507.03607.
- **C. Bonhomme**, A. Dulaunoy, *Scoring Vulnerabilities by Leveraging Activity Data from the Fediverse*. FIRST Cyber Threat Intelligence Conference 2025, Berlin, Germany, April 2025.

2024

- **C. Bonhomme**, A. Dulaunoy, *Correlating Security Advisories with Vulnerability-Lookup*. CSAF Community Days 2024, Munich, Germany, December 2024.
- **C. Bonhomme**, *Improving the CVD Process*. Hack.lu 2024, Luxembourg, October 2024.
- **C. Bonhomme**, *Vulnerability-Lookup – An Open Source Tool to Support CVD Processes*. Valencia 2024 UNDP/UNICC/FIRST Technical Colloquium, Valencia, Spain, October 2024.

2018

- F. Mathey, **C. Bonhomme**, J. Rocha, J. Lombardi, B. Joly, *Risk Assessment Optimisation with MONARC*. Hack.lu 2018, Luxembourg, October 2018.

2012

- **C. Bonhomme**, G. Arnould, D. Khadraoui, *Dynamic Carpooling Mobility Services based on Secure Multi-Agent Platform*. Global Information Infrastructure and Networking Symposium (GIIS'12), Choroní, Venezuela, December 2012.

2011

- M. Ouedraogo, H. Mouratidis, A. Hecker, **C. Bonhomme**, D. Khadraoui, D. Preston, *A New Approach to Evaluating Security Assurance*. 7th International Conference on Information Assurance and Security (IAS 2011), Melaka, Malaysia, December 2011.
- G. Guemkam, C. Feltus, P. Schmitt, **C. Bonhomme**, D. Khadraoui, Z. Guessoum, *Reputation Based Dynamic Responsibility to Agent Assignment for Critical Infrastructure*. IEEE/WIC/ACM International Conference on Intelligent Agent Technology (IAT 2011), pp. 272–275.
- **C. Bonhomme**, C. Feltus, M. Petit, *Dynamic Responsibilities Assignment in Critical Electronic Institutions – A Context-Aware Solution for in Crisis Access Right Management*. Sixth International Conference on Availability, Reliability and Security

(ARES 2011), Vienna, Austria.

- G. Guemkam, C. Feltus, **C. Bonhomme**, P. Schmitt, B. Gâteau, D. Khadraoui, Z. Guessoum, *Financial Critical Infrastructure: A MAS Trusted Architecture for Alert Detection and Authenticated Transactions*. Sixth IEEE Conference on Network Architectures and Information Systems Security (SAR-SSI 2011), La Rochelle, France.

2010

- T. Schaberreiter, **C. Bonhomme**, J. Aubert, C. Incoul, *Support Tool Development for Real-Time Risk Prediction in Interdependent Critical Infrastructures*. International Workshop on Risk and Trust in Extended Enterprises (RTEE'2010), San Jose, CA, USA, November 2010.
- **C. Bonhomme**, C. Feltus, D. Khadraoui, *A Multi-Agent Based Decision Mechanism for Incident Reaction in Telecommunication Network*. Poster, 8th ACS/IEEE International Conference on Computer Systems and Applications (AICCSA 2010), Hammamet, Tunisia.
- P. Schmitt, **C. Bonhomme**, J. Aubert, B. Gâteau, *Programming Electronic Institutions with UTOPIA*. Demo session, 22nd International Conference on Advanced Information Systems Engineering (CAiSE'10), 2010.
- P. Schmitt, **C. Bonhomme**, B. Gâteau, *Easy Programming of Agent-Based Electronic Institution with UTOPIA*. 10th International Conference on New Technologies of Distributed Systems (NOTERE 2010), Tozeur, Tunisia, May–June 2010.
- C. Feltus, D. Khadraoui, **C. Bonhomme**, *Electric Blackout Prevention: Toward a Computer-Mediated Weather Alert Broadcasting Solution*. International Conference on Society and Information Technologies (ICSIT 2010), Orlando, FL, USA, pp. 45–50.

2008

- **C. Bonhomme**, J.-C. Bettinger, *Les paysages fractals*. Université Paul Verlaine, Metz, France, 2008 (in French).

Last updated: November 6, 2026